Junyu Tang

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Academic websites: Google Scholar
Office Location: East 501, Physics building, PKU

Research Interests

Strongly correlated systems: Superconductivity, non-Abelian anyon, Chiral spin liquid, Mott insulators
Spintronics and Quantum magnetism: Spin-orbit torques, antiferromagnetic dynamics, spin-lattice coupling, flavor-wave theory
Topology in condensed matter: Topological magnetic materials, topological magnon

Education

2025–present: Postdoc at Peking University
International Center for Quantum Materials (Collaboration advisor: Gang v. Chen)
2020–2025: Ph.D. in Physics
University of California, Riverside
Dissertation: *Intrinsic Spin-Orbit Torque in Magnetic Topological Insulators* (Advisors: Ran Cheng).
2015–2019: B.S. in Applied Physics
University of Science and Technology of China
Thesis: *DFT calculations on 2D ferromagnetic half-metal* (Advisors: Ping Cui and Zhenyu Zhang)

Honors and Awards

2025	National Postdoctoral Overseas Talent Recruitment Program of China (Category I), Ministry of Education of China
2025	Boya Postdoctoral Fellowship, Peking University
2025	Dissertation Completion Fellowship Award, UCR
2020	Dean's Distinguished Fellowship, UCR
2018	Excellent Student Scholarship, USTC
2017	First Prize, 13th University Physics Research Experiment Competition, USTC
2016	Second Prize, Optics Research Competition, USTC

Professional Experiences

2026 Jul.
International Workshop, Max Planck Institute for Solid State Research, Stuttgart, Germany
2024 Jul.
Princeton Summer School on Condensed Matter Physics, Institute for Advanced Study, Princeton.
2023 Jul.
Principles and Applications of Symmetry in Magnetism Summer School, Colorado State University.
2021 Jan. – 2025 Mar.
Research Assistant, University of California, Riverside.
Winter & Fall 2020
Teaching Assistant, University of California, Riverside.
2019. Jul. – 2019 Dec.
Research Assistant, International Center for Quantum Design of Functional Materials (ICQD), USTC
2019 Jan. – 2019 Jun.
Research Assistant, Institute for Advanced Study, Shenzhen University.
2018 Jul. – 2018 Aug.
Summer Research at the University of Western Australia. Advisor: Mikhail Kostylev.
2017 Aug.
Summer school at Magdalene & Oriel college, University of Cambridge & Oxford, UK.

Publications

Published/Accepted:

- **Junyu Tang**, Hong-Hao Song, and Gang v. Chen, "*Phononic enhancement and detection of hidden spin-nematicity and dynamics in quantum magnets*", Physical Review B **114**, 014407 (2026)
- **Junyu Tang** and Ran Cheng, "*Proper Theory of Magnon Orbital Angular Momentum at Equilibrium*", Physical Review Letters **136**, 236702 (2026)
- Gregory Fritjofson, **Junyu Tang**, Atul Regmi, Jacob Hanson-Flores, Justin Michel, Fengyuan Yang, Ran Cheng, and Enrique Del Barco, "*Coherent Spin Pumping Originated from Sub-Terahertz Néel Vector Dynamics in Easy-Plane α-Fe_2O_3/Pt*", Physical Review Letters **135**, 216704 (2025)
- **Junyu Tang**, Hantao Zhang, and Ran Cheng, "*Néel spin-orbit torque in antiferromagnetic quantum spin and anomalous Hall insulators*", Nature Communications **16**, 7790 (2025)
- Josiah Keagy, Haoyu Liu, **Junyu Tang**, Weilun Tan, Wei Yuan, Sumukh Mahesh, Ran Cheng, and Jing Shi, "*Disorder-induced suppression of antiferromagnetic magnon transport in Cr_2O_3*", Physical Review Applied **24**, L011005 (2025)
- Gregory Fritjofson, Atul Regmi, Jacob Hanson-Flores, Justin Michel, **Junyu Tang**, Fengyuan Yang, Ran Cheng, and Enrique Del Barco, "*Probing polarization-tunable sub-terahertz spin pumping in bimodal α-Fe_2O_3/Pt*", npj Spintronics **3**, 24 (2025)
- I-Hsuan Kao, **Junyu Tang**, Gabriel Calderón Ortiz, Menglin Zhu, Sean Yuan, Rahul Rao, Jiahan Li, James H. Edgar, Jiaqiang Yan, David G. Mandrus, Kenji Watanabe, Takashi Taniguchi, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh, "*Unconventional unidirectional magnetoresistance in heterostructures of a topological semimetal and a ferromagnet*", Nature Materials **24**, 1049–1057 (2025)
- Yang Cheng, Hanshen Huang, **Junyu Tang**, Joseph Lanier, Katelyn Lazareno, Hao-Kai Chang, Shang-Jui Chui, Chao-Yao Yang, Fengyuan Yang, Ran Cheng, and Kang L. Wang, "*Observation of Real-Time Spin-Orbit Torque Driven Dynamics in Antiferromagnetic Thin Film*", Advanced Materials **37**, 2417240 (2025)
- **Junyu Tang** and Ran Cheng, "*Lossless Spin-Orbit Torque in Antiferromagnetic Topological Insulator $MnBi_2Te_4$*", Physical Review Letters **132**, 136701 (2024)
- Yang Cheng, **Junyu Tang**, Justin J. Michel, Su Kong Chong, Fengyuan Yang, Ran Cheng, and Kang L. Wang, "*Unidirectional Spin Hall Magnetoresistance in Antiferromagnetic Heterostructures*", Physical Review Letters **130**, 086703 (2023)
- **Junyu Tang** and Ran Cheng, "*Absence of cross-sublattice spin pumping and spin-transfer torques in collinear antiferromagnets*", APL Materials **11**, 111117 (2023)
- **Junyu Tang** and Ran Cheng, "*Voltage-driven exchange resonance achieving 100% mechanical efficiency*", Physical Review B **106**, 054418 (2022)
- Sheng Yu, **Junyu Tang**, Yu Wang, Feixiang Xu, Xiaoguang Li, and Xinzhong Wang, "*Recent advances in two-dimensional ferromagnetism: strain-, doping-, structural- and electric field-engineering toward spintronic applications*", Science and Technology of Advanced Materials **23**, 140–160 (2022)

Preprints:

- **Junyu Tang** and Gang v. Chen, "*Non-Abelian Anyon Braiding with Quantum-Antidot Interferometry*", arXiv:2606.24930 (2026)

Scientific Presentations

Contributed Talk:

2025 Jan: **2025 Joint MMM-Intermag Conference, New Orleans**
Néel Spin Torque in Antiferromagnetic Quantum Spin and Anomalous Hall Insulators

2024 Mar: **APS March Meeting, Minneapolis**
Antiferromagnetic Quantum Spin and Anomalous Hall Insulator: Dissipationless Neel SOT

2023 Nov: **Annual Conference on Magnetism and Magnetic Materials, Dallas**
Exotic Spin-Orbit Torque in Antiferromagnetic Topological Insulator $MnBi_2Te_4$

2023 Mar: **APS March Meeting, Las Vegas**
High-Efficient Spin-Orbit Torque in Antiferromagnetic Topological Insulator $MnBi_2Te_4$

2023 Mar: **IEEE Magnetic Frontiers Quantum Technology Conference, Orlando**
Absence of cross-sublattice spin torques and spin pumping in collinear AFM

2022 Mar: **APS March Meeting, Chicago**
Anomalous Exchange Resonance Driven by SOT in few SL $MnBi_2Te_4$

2022 Jan: **2022 Joint MMM-INTERMAG Conference, New Orleans**
Anomalous Exchange Resonance Driven by SOT in Intrinsic AFM Topological Insulator

Poster:

2026 Jul: **International Workshop on New Insights into Quantum Materials, MPI-FKF**
Non-abelian Anyon Braiding with Quantum Anti-dot Interferometry

2022 Oct: **Annual Conference on Magnetism and Magnetic Materials, Minneapolis**
Cross-sublattice spin torques and spin pumping in antiferromagnets

2019 Sep: **Chinese Physical Society Fall Meeting, Zhengzhou**
Two-dimensional half-metallic ferromagnets: $FeCl_2$ *and* $MnSe_2$

Peer Review Experience

- **Journal: Physical Review Letters**
  Reviewed articles on electronic and magnonic transport in topological materials and altermagnet.
- **Journal: Physical Review B**
  Reviewed articles related to spin-orbit torques and nonHermitian system.
- **Journal: Nature Communications**
  Evaluated manuscripts related with excitation of antiferromagnetic magnon.
- **Journal: Applied Physics Letters**
  Evaluated research papers on magnon transport in antiferromagnets.

Teaching Experience

- GENERAL PHYSICS LABORATORY 02LB (015), Fall 2020, UCR
- GENERAL PHYSICS LABORATORY 02LB (007), Fall 2020, UCR
- GENERAL PHYSICS LABORATORY 02LB (034), Winter 2020, UCR
- GENERAL PHYSICS LABORATORY 02LB (013), Winter 2020, UCR

Scientific Skills

Programming:

Python, C++, MATLAB, Wolfram Mathematica

Computational:

KWANT, Mumax3, Quantum ESPRESSO, VASP

AI Tools:

ChatGPT, Codex, Claude, Copilot, Deepseek, Gemini